

5(a)	A photon is a quantum of electromagnetic radiation whose energy is given by the product of the Planck's constant and its frequency.
(b)(i)	$p = \frac{h}{\lambda}$ $= \frac{6.63 \times 10^{-34}}{656 \times 10^{-9}}$ $= 1.01 \times 10^{-27} \text{ kg m s}^{-1}$
(ii)	$\Delta x = c \Delta t$ $= (3.00 \times 10^8)(9.00 \times 10^{-15})$ $= 2.70 \times 10^{-6} \text{ m}$
	$\Delta x \Delta p \gtrsim h$ $\text{minimum } \Delta p \approx \frac{h}{\Delta x}$ $= \frac{6.63 \times 10^{-34}}{2.70 \times 10^{-6}}$ $= 2.46 \times 10^{-28} \text{ kg m s}^{-1}$
(c)	When isolated gas atoms are in the excited state, electrons occupying the higher energy levels will spontaneously transit from the higher energy levels to lower energy levels.
	Each transition produces a photon whose energy is equal to the energy difference of the two electron energy levels involved in the transition.
	Discontinuous/discrete coloured lines are produced on a dark background implying only photons of specific energy/wavelength/frequency is emitted, indicating that the energy levels in an isolated atom are discrete.
(d)	$\Delta E = hf$ $= (6.63 \times 10^{-34})(3.19 \times 10^{15})$ $= 2.115 \times 10^{-18} \text{ J}$ $= \frac{2.115 \times 10^{-18}}{1.60 \times 10^{-19}} \text{ eV}$ $= 13.22 \text{ eV}$
	$E_{\text{higher}} - E_{\text{lower}} = 13.22$ $E_{\text{higher}} = 13.22 + (-13.6)$ $= -0.38 \text{ eV}$
(e)(i)	$\frac{hc}{\lambda} = \phi + eV_s$
(ii)	$V_s = \frac{hc}{e} \left(\frac{1}{\lambda} \right) - \frac{\phi}{e}$ Plotting V_s against $\frac{1}{\lambda}$, $\frac{hc}{e}$ is equal to the gradient.
	$\text{gradient of graph} = \frac{1.80 - 0.70}{(3.00 - 2.10) \times 10^6}$ $= 1.222 \times 10^{-6} \text{ V m}$

	$\frac{hc}{e} = 1.222 \times 10^{-6}$ $h = \frac{(1.222 \times 10^{-6})(1.60 \times 10^{-19})}{3.00 \times 10^8}$ $= 6.52 \times 10^{-34} \text{ J s}$
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